SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	25 June 2024
Team ID	
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	Develop an intelligent agricultural recommendation system that uses machine learning to predict the most suitable crop based on soil nutrients and environmental conditions, improving farming productivity and resource efficiency
Scope	The project analyzes NPK levels, temperature, humidity, pH, and rainfall data using ML algorithms (KNN, Logistic Regression, K-Means Clustering) and provides real-time crop recommendations through a Flask web application
Problem Statement	
Description	Farmers face challenges in selecting optimal crops due to lack of scientific analysis tools, leading to crop failure, resource wastage, and financial losses
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Proposed Solution	
Approach	Collect agricultural dataset → Preprocess data → Train multiple ML models → Evaluate performance → Deploy best model via Flask web app
Key Features	• ML-based crop prediction using 7 environmental parameters • Interactive web interface with prediction form • Prediction history management (view/delete) • Multiple model comparison (KNN, Logistic Regression, K-Means) • Real-time crop recommendation

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications	Intel i5 or higher / Any modern CPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	500 MB (dataset + models + app)
Software		
Frameworks	Python frameworks	Flask
Libraries	ML & data libraries	scikit-learn, pandas, NumPy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, PyCharm
Data		
Data	Source, size, format	Kaggle Crop_recommendation .csv, 2200 rows, CSV